Multi-Speaker Video Dubbing Runbook

Zero-Shot Cross-Lingual Pipeline Operation & Deployment Guide

HARDWARE TARGET
NVIDIA RTX 3090 (WSL2 / Ubuntu)

ENVIRONMENT
Python 3.10 | CUDA 12.1 | PyTorch 2.5.1

CORE MODELS
Whisper-large-v3, NLLB-200, XTTS-v2

Pipeline Execution Steps

1. Environment Activation & Dependency Verification

Stage 0

Ensure you are inside the virtual environment with pinned dependencies (NumPy 1.26.4):

```
cd /mnt/d/aipipeline/stage1
source venv/bin/activate
```

2. Transcription & Smart Diarization

Stage 1

Transcribe the source video into Japanese segments with dynamic speaker rotation:

```
python3 export_diarization.py
```

- **Input:** Source video (`my_video.mp4`)
- **Output:** `segments_diarized.json`

3. Target Speaker Reference Extraction

Stage 1.5

Extract clean 10-second audio clips for each unique speaker straight from the video:

```
python3 extract_voices.py
```

- **Output:** `speaker_refs/SPEAKER_00.wav`, `SPEAKER_01.wav`, etc.

4. Voice-Cloned TTS Synthesis & Translation

Stage 2

Translate Japanese text via NLLB-200 and synthesize English audio using cloned XTTS-v2 voices:

```
python3 stage2_diarized_tts.py
```

- **Output:** Time-synced audio track (`audio_en_synced.wav`)

5. Final Audio-Video Muxing

Stage 3

Combine the original video track, ducked background audio, and new dubbed speech:

```
python3 stage3_assemble.py
```

- **Output:** `final_dubbed_video.mp4`

Environment Backup & Deployment

Deployment via Requirements Manifest

Git & Replicability

Export the locked package versions to allow clean deployment on secondary machines:

```
pip freeze > requirements.txt
git add requirements.txt
git commit -m "Update environment requirements manifest"
git push origin main
```